

COMPAS Model Audit Report

Responsible Machine Learning Audit

Sowmya Vara

To the Instructor and Stakeholders of the COMPAS Model Evaluation:

Report on the Model Evaluation

I conducted an audit of Logistic Regression (LR) and Gradient Boosted Tree (GBT) models trained on the COMPAS dataset. The audit evaluates robustness, fairness, and privacy risks under adversarial conditions, including evasion attacks, data poisoning, and membership inference. The analysis focuses on adversarial perturbations, fairness degradation under label manipulation, and potential privacy leakage through inference attacks.

Audit Procedures and Findings

1. Adversarial Evasion (PGD Attack)

Both models were evaluated under a Projected Gradient Descent (PGD) evasion attack across perturbation levels ($\epsilon \in \{0.25, 0.5, 1.0, 2.0\}$). Adversarial examples were generated using LR and transferred to GBT.

Findings:

Both models exhibit increasing false positive rates as perturbation strength increases. The Adverse Impact Ratio (AIR) approaches parity (≈ 1.0) as ϵ increases; however, this is not a meaningful fairness improvement. Instead, both groups experience uniformly high error rates ($FPR \approx 1.0$), indicating a collapse in model utility. Both models show similar vulnerability under attack, with differences primarily attributable to transfer effects rather than inherent robustness. These results highlight that apparent fairness improvements under adversarial conditions can be misleading when overall performance degrades.

2. Data Poisoning and Fairness Degradation

A label-flip poisoning attack was conducted targeting demographic groups to assess fairness vulnerability.

Findings:

The model exhibits a significant vulnerability in the form of a stealth zone, where fairness degrades substantially without affecting performance metrics. Across poisoning rates from 0.02 to 0.30, AUC remains stable (≈ 0.7345 to ≈ 0.7311), while AIR decreases significantly (≈ 0.50 to ≈ 0.33) when targeting African-American defendants. This demonstrates that fairness can be systematically degraded without triggering performance-based alerts. In particular, accuracy-based monitoring fails completely to detect this failure mode, as model utility appears unchanged despite substantial fairness degradation.

3. Drift Detection Evaluation

The effectiveness of the Population Stability Index (PSI) was evaluated for detecting poisoning attacks.

Findings:

PSI fails to detect the attack, with values approximately equal to 0.0 across all features. Since the poisoning attack modifies labels without altering feature distributions, feature-based drift monitoring remains insensitive. This highlights a critical limitation: monitoring input distributions alone is insufficient for detecting attacks that target the relationship between features and labels.

4. Membership Inference Risk

Privacy risk was assessed using a shadow-model membership inference (MI) attack.

Findings:

Both models exhibit MI AUC values near 0.50 (LR $\approx$ 0.497, GBT $\approx$ 0.506), indicating that the attack performs no better than random guessing. Although GBT shows a larger generalization gap, this does not translate into meaningful privacy leakage. Varying L2 regularization strength has negligible impact on MI AUC, suggesting no meaningful privacy–utility tradeoff in this setting. These results indicate that membership inference risk is minimal under the tested conditions.

Opinion

Based on the audit procedures performed, I conclude that:

- The models demonstrate acceptable performance under standard evaluation but are not robust to adversarial and data integrity threats.
- Fairness is highly vulnerable to manipulation through label-flip poisoning, and this vulnerability is not detectable using standard performance or drift monitoring techniques.
- Privacy risks from membership inference are minimal in this setting.

Overall, while the models appear reliable under traditional evaluation metrics, they are not suitable for high-stakes deployment without additional safeguards, particularly for fairness monitoring and data integrity protection.

Recommendations

To address the identified risks, I recommend:

- Implementing strict data governance and access controls to prevent label manipulation
- Deploying continuous subgroup fairness monitoring (e.g., AIR, group-wise FPR) with emphasis on tracking changes over time
- Augmenting monitoring systems to detect anomalies in label distributions and group-wise error rates
- Incorporating adversarial testing into the model validation lifecycle

Limitations of the Audit

This audit is based on the available dataset and experimental setup. Results may vary under different data distributions, model configurations, or attack strategies. The conclusions reflect observed behavior under the tested conditions and should be validated under real-world deployment scenarios.

Conclusion

This audit highlights a critical gap between traditional model evaluation and real-world robustness. While performance and privacy risks appear controlled, fairness can be significantly degraded without detection. Addressing these vulnerabilities requires integrating fairness-aware monitoring, secure data practices, and continuous auditing into the model lifecycle.